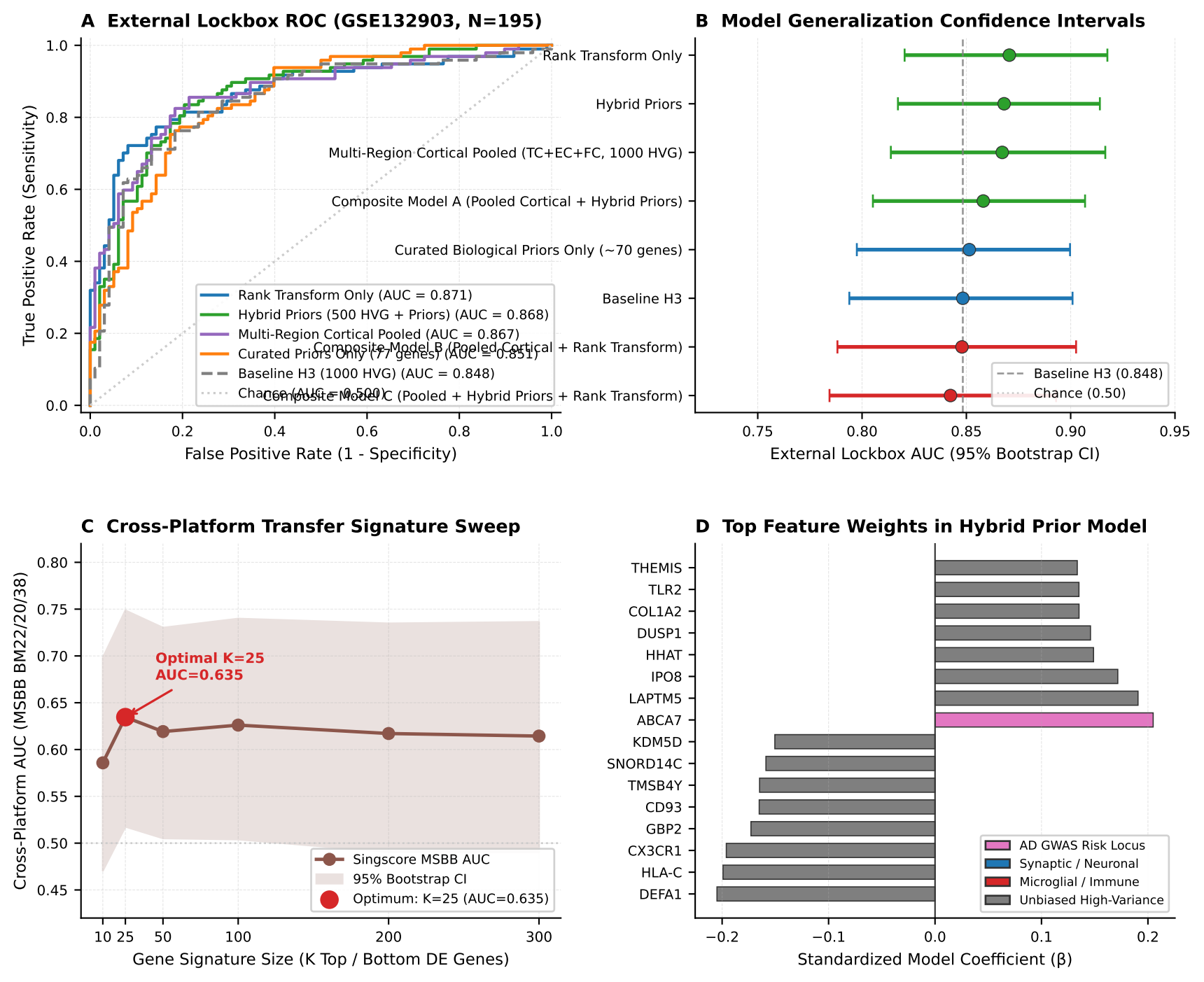
A External Lockbox ROC (GSE132903, N=195)
True Positive Rate (Sensitivity)
False Positive Rate (1 - Specificity)
Rank Transform Only (AUC = 0.871)
Hybrid Priors (500 HVG + Priors) (AUC = 0.868)
Multi-Region Cortical Pooled (AUC = 0.867)
Curated Priors Only (77 genes) (AUC = 0.851)
Baseline H3 (1000 HVG) (AUC = 0.848)
Chance (AUC = 0.500)
B Model Generalization Confidence Intervals
Rank Transform Only
Hybrid Priors
Multi-Region Cortical Pooled (TC+EC+FC, 1000 HVG)
Composite Model A (Pooled Cortical + Hybrid Priors)
Curated Biological Priors Only (~70 genes)
Baseline H3
Composite Model B (Pooled Cortical + Rank Transform)
Composite Model C (Pooled + Hybrid Priors + Rank Transform)
Baseline H3 (0.848)
Chance (0.50)
External Lockbox AUC (95% Bootstrap CI)
C Cross-Platform Transfer Signature Sweep
Cross-Platform AUC (MSBB BM22/20/38)
Optimal K=25
AUC=0.635
Singscore MSBB AUC
95% Bootstrap CI
Optimum: K=25 (AUC=0.635)
Gene Signature Size (K Top / Bottom DE Genes)
D Top Feature Weights in Hybrid Prior Model
THEMIS
TLR2
COL1A2
DUSP1
HHAT
IPO8
LAPTM5
ABCA7
KDM5D
SNORD14C
TMSB4Y
CD93
GBP2
CX3CR1
HLA-C
DEFA1
AD GWAS Risk Locus
Synaptic / Neuronal
Microglial / Immune
Unbiased High-Variance
Standardized Model Coefficient (β)